You've now seen some ways PgHero can be used as a comprehensive PostgreSQL Performance dashboard for your team. PgHero is so useful that you'll continue to work with it in upcoming chapters. For now, you'll focus in on Query Performance.

What other query performance PostgreSQL level visibility tools are available?

Once you have broad query level observability, you're prepared to identify individual queries to optimize. To optimize queries in PostgreSQL, you'll need to see how PostgreSQL plans to execute the query. To do that you'll want the EXPLAIN tool.

Analyzing Query Execution Plans

SQL is a declarative language. When you write a SQL statement you declare the results you want and PostgreSQL determines the best way to get them. PostgreSQL shows you a lot of detail about how it plans and estimates what it will do.

The steps PostgreSQL takes are called the Query Execution Plan. Generating and reading Query Execution Plans is critical to understanding queries, identifying issues, and making improvements. This section will introduce or act as a refresher on how to generate and read query execution plans.

Open up a psql prompt and add EXPLAIN in front of any SQL query. Run this example:

```
EXPLAIN SELECT 1;
                QUERY PLAN
------------------------------------------
 Result  (cost=0.00..0.01 rows=1 width=4)
(1 row)
```

EXPLAIN is a tool with many more arguments like ANALYZE, BUFFERS, and FORMAT that you'll see how to use. The main purpose is to show some details about the plan it has in mind to fulfill your request.

The PostgreSQL query execution planner is also called the *optimizer*. This name is based on how it works. The planner will compare multiple query execution options against each other when that's possible, and choose the optimal plan.

Here are some basics about planning. The planner creates plans broken into nodes and each node has a cost. The costs are totaled up and the lowest cost plan is selected. This planning happens at runtime when you run your query and does add a bit of latency to the process. The planning time itself is added